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(54) **PHOTOVOLTAIC MODULES WITH OFFSET LAYERS**

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See application file for complete search history.

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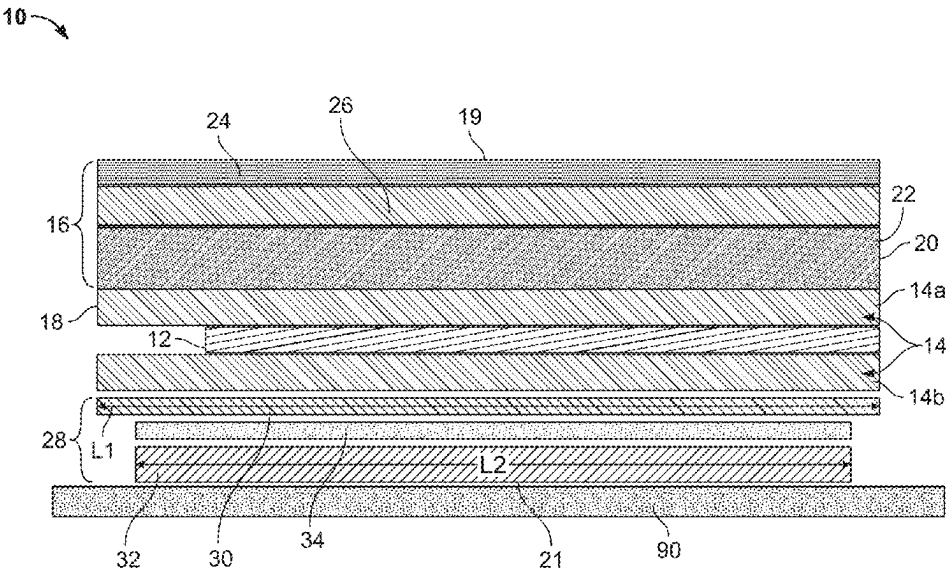
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(57) **ABSTRACT**

A system includes a plurality of photovoltaic modules installed on a roof deck. Each of the photovoltaic modules includes solar cells, a first encapsulant encapsulating the solar cells, a frontsheet and a backsheet. The front sheet includes a glass layer. The backsheet includes a first layer and a second layer. Each of the first and second layers includes first, second, third and fourth edges. The first edge of the first layer is offset from the first edge of the second layer, the second edge of the first layer is offset from the second edge of the second layer, the third edge of the first layer is offset from the third edge of the second layer, and the fourth edge of the first layer is offset from the fourth edge of the second layer.

19 Claims, 6 Drawing Sheets



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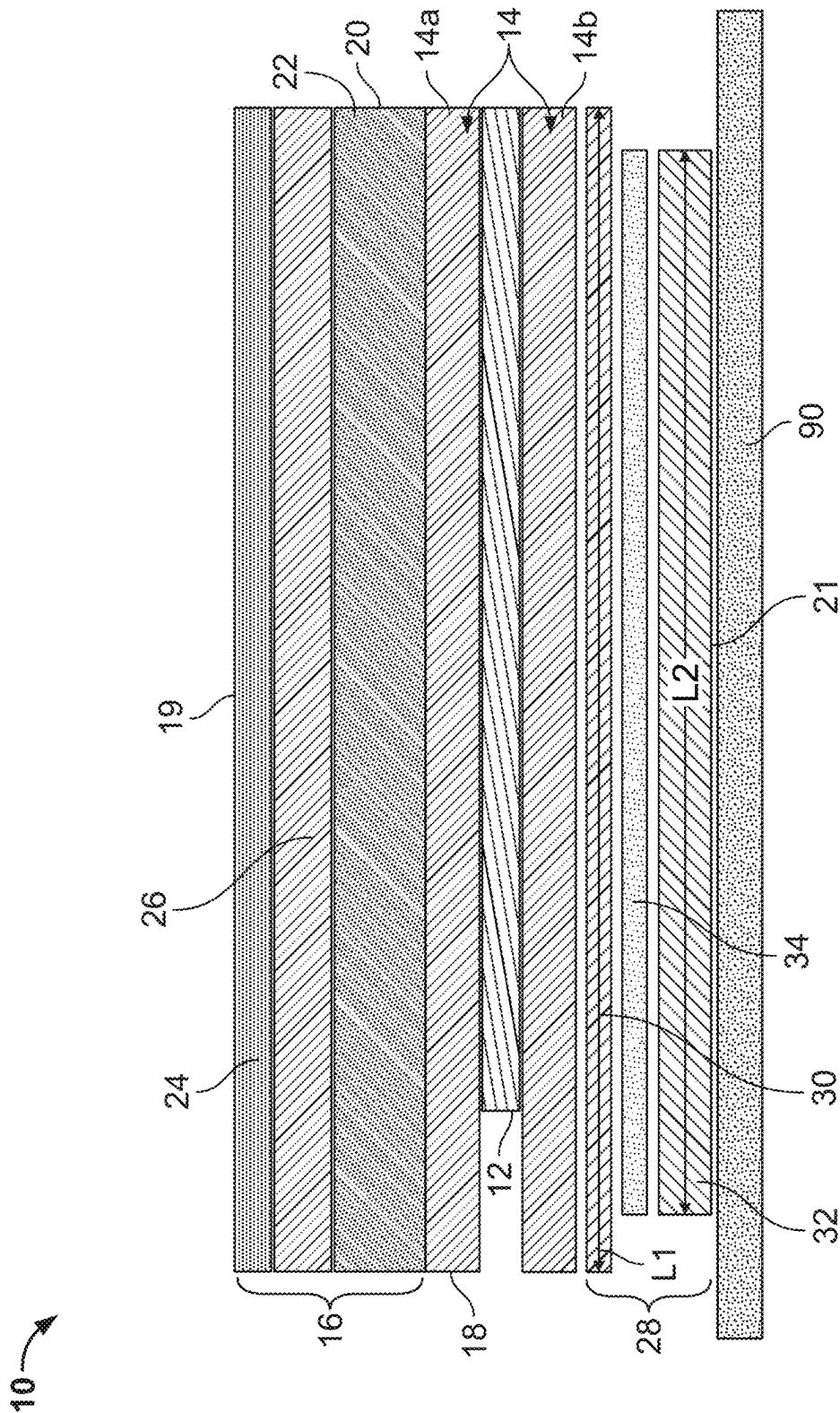


FIG. 1

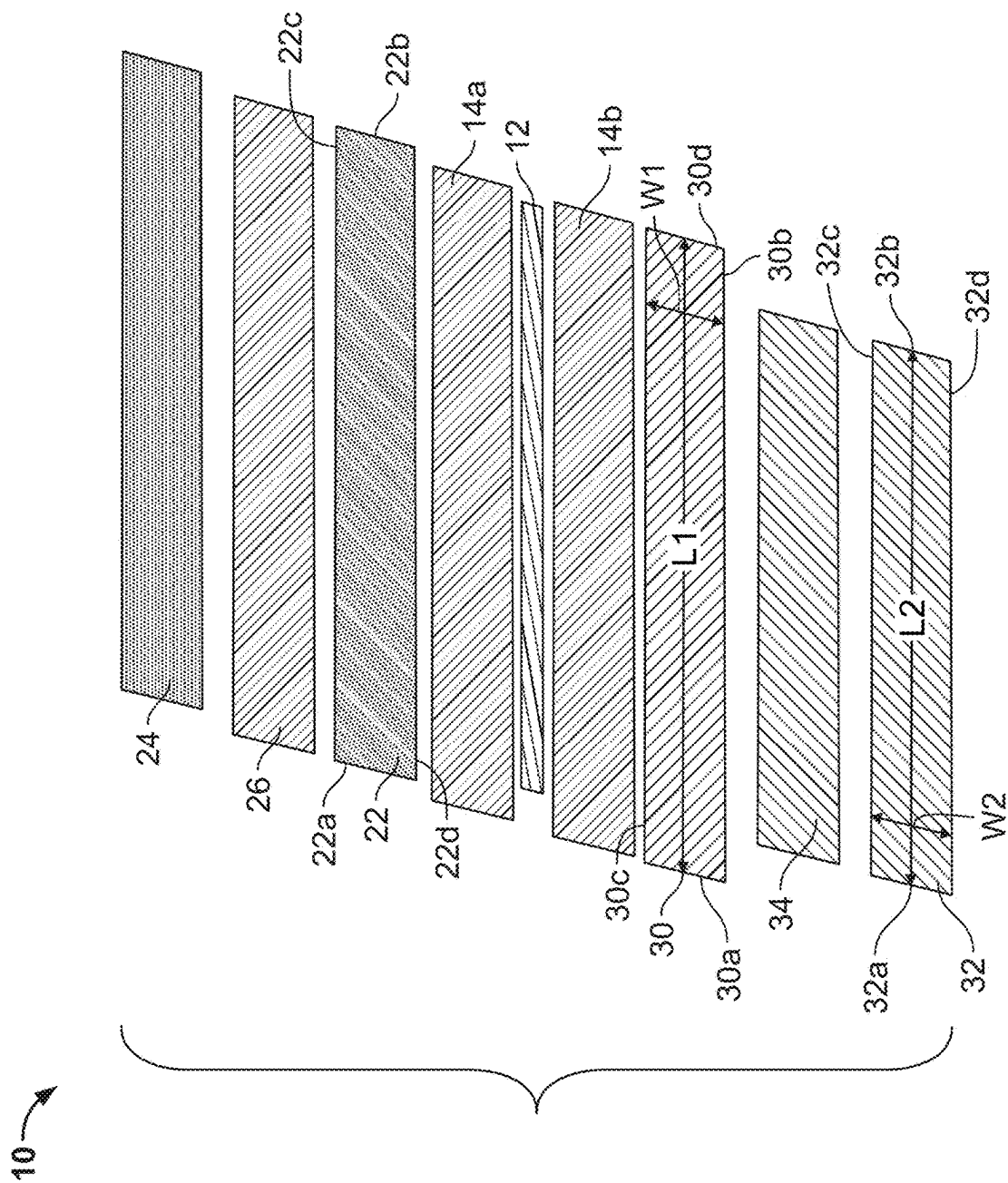


FIG. 2

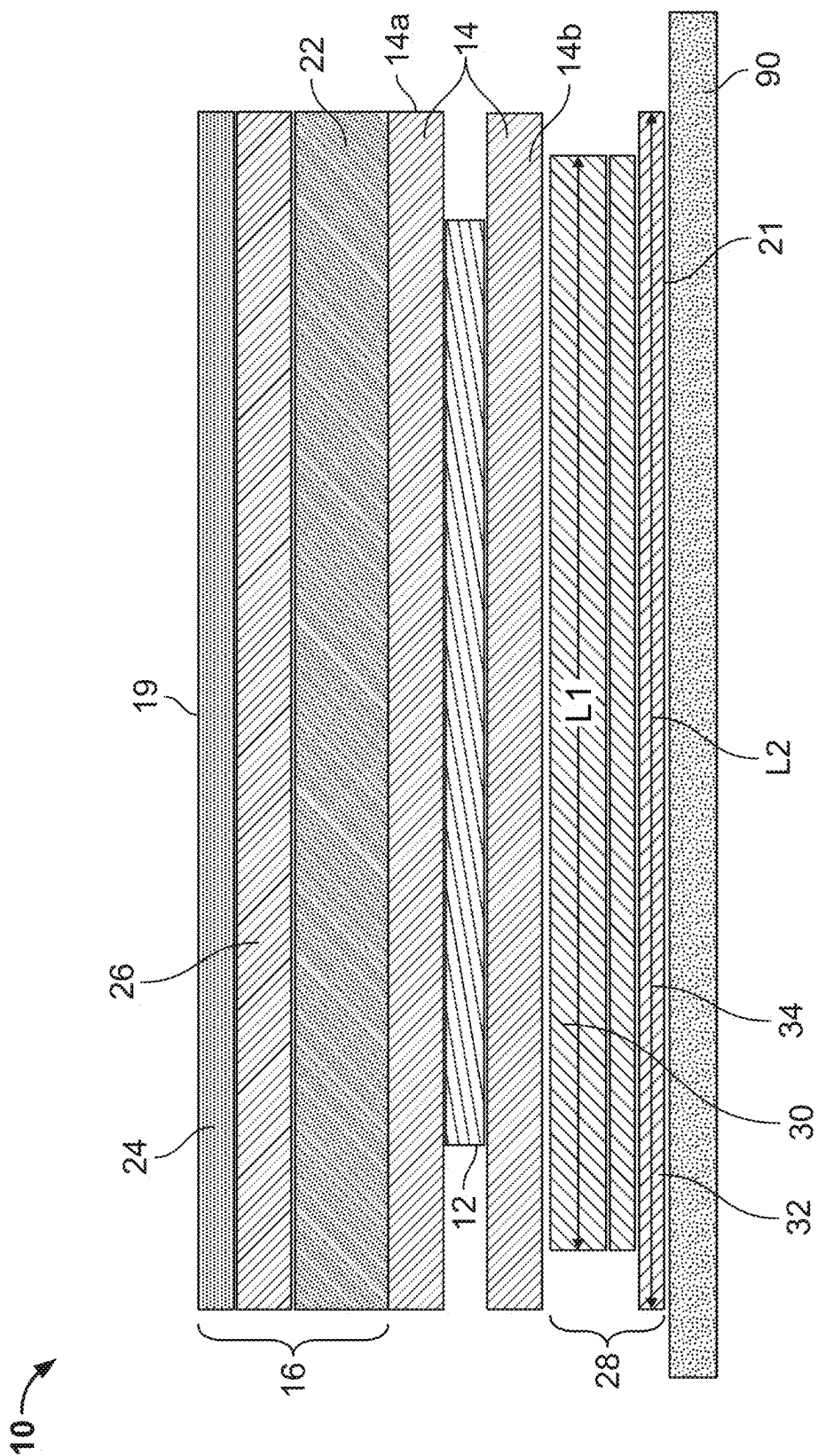
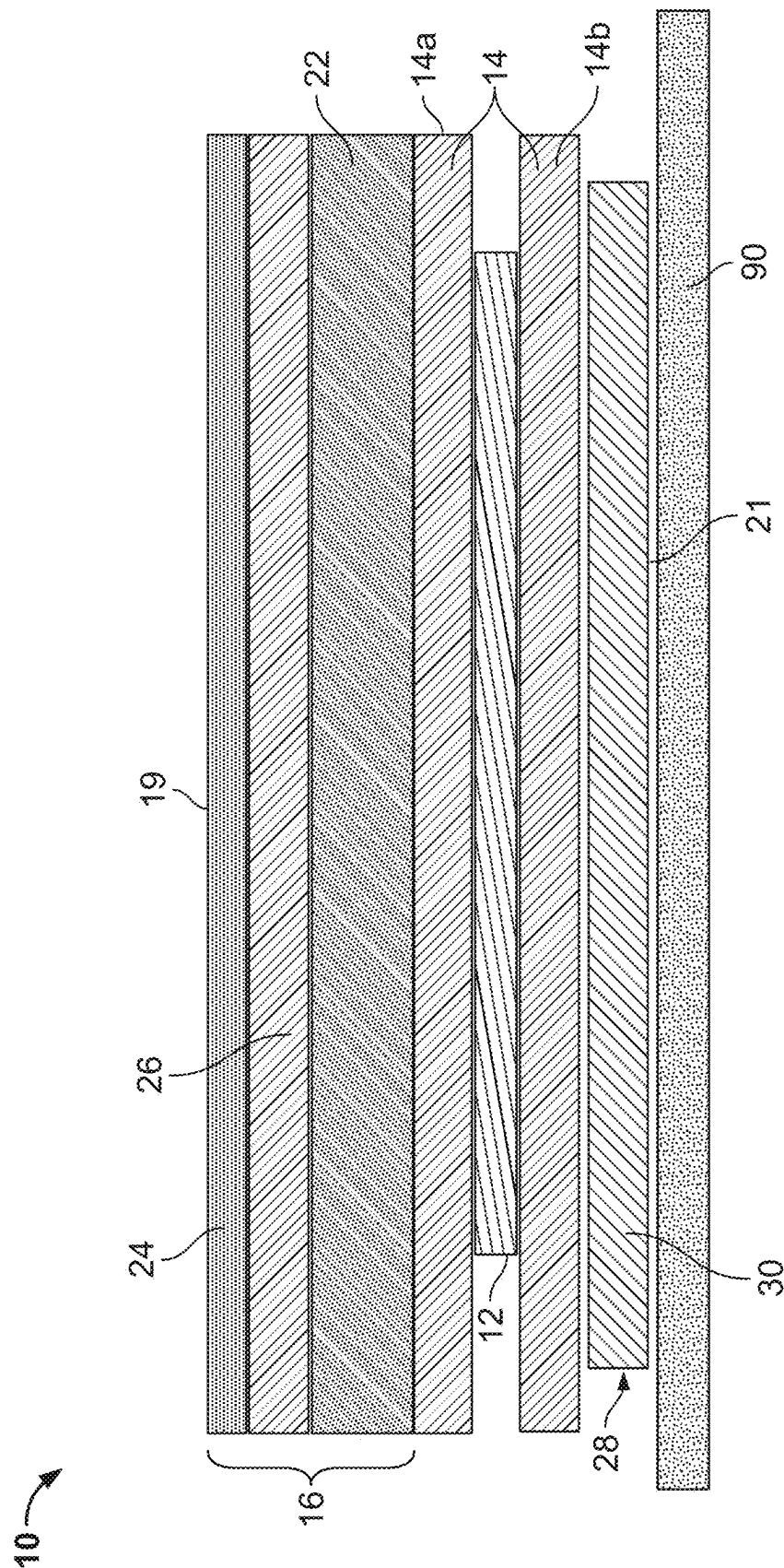


FIG. 3



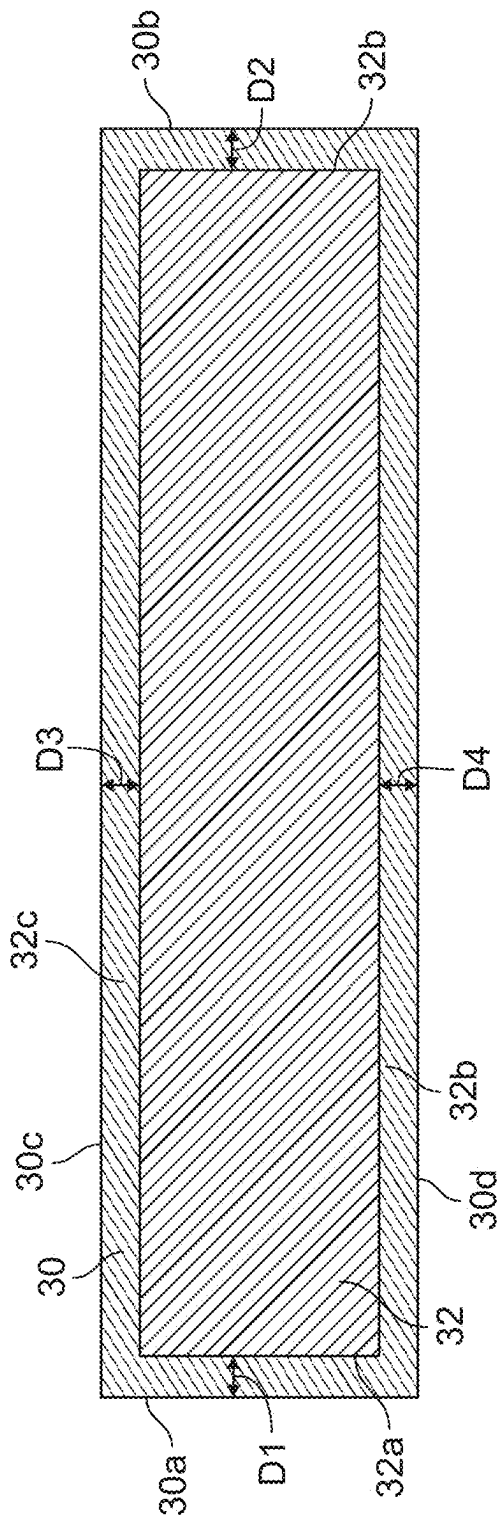


FIG. 5

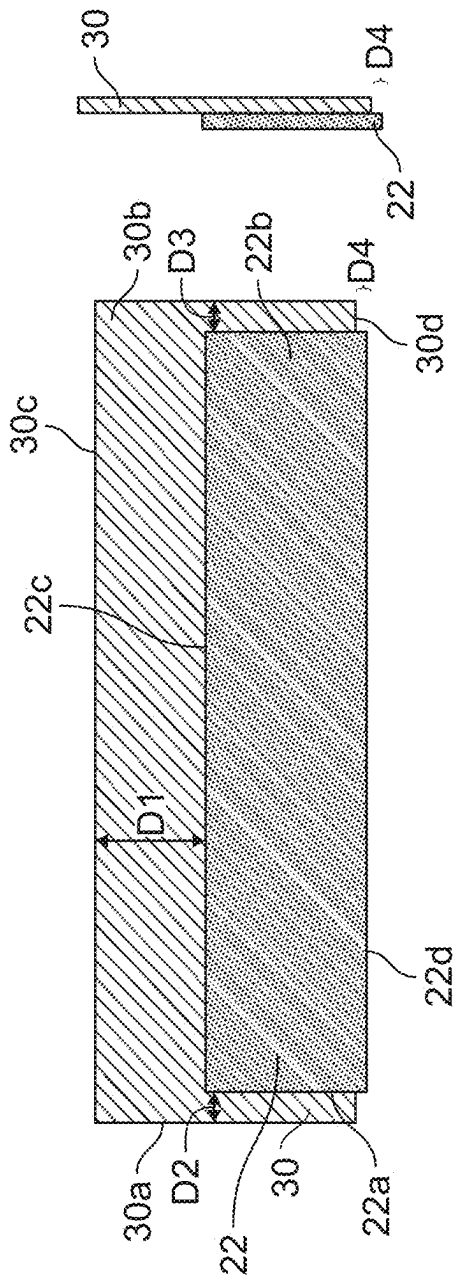


FIG. 6

FIG. 7

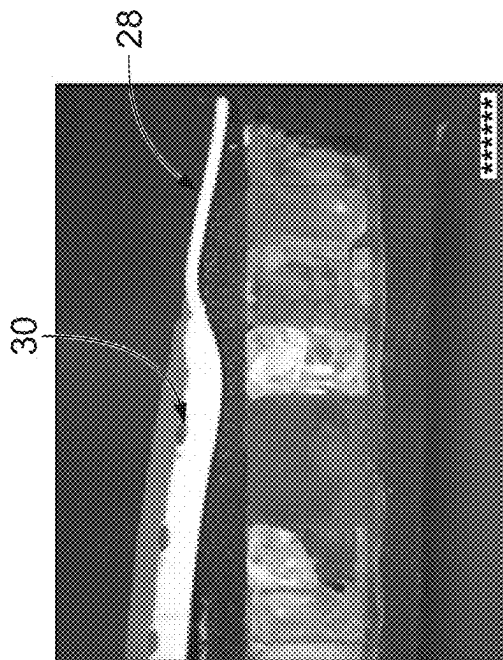
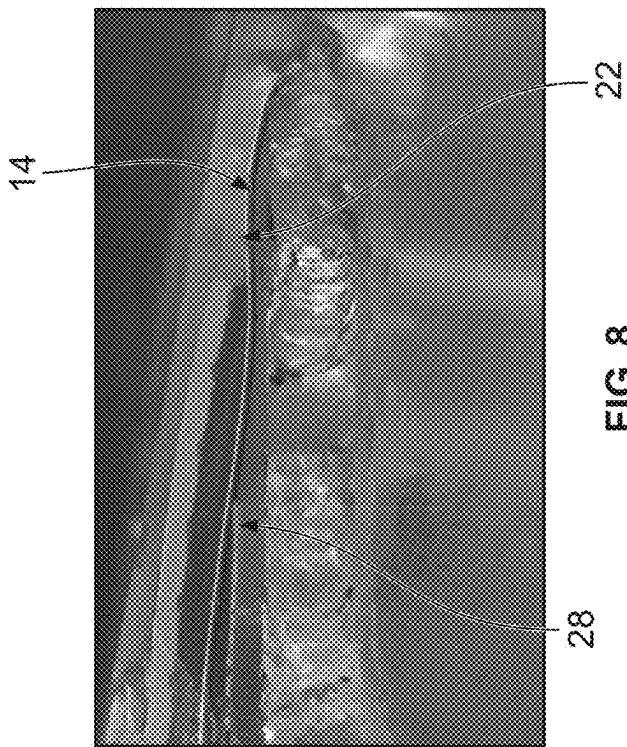


FIG. 9

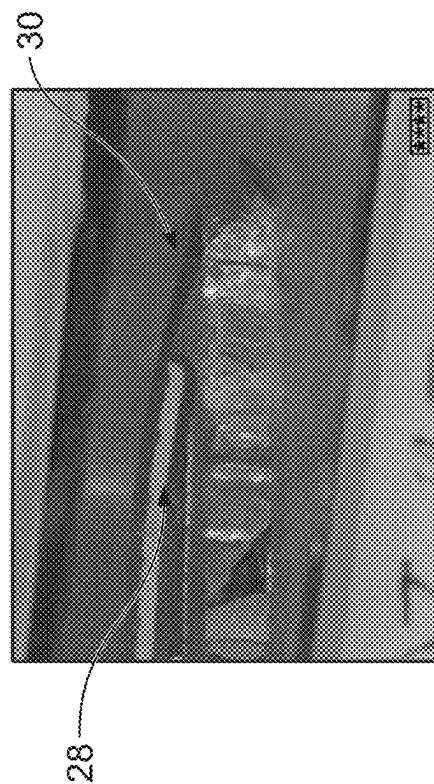


FIG. 10

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PHOTOVOLTAIC MODULES WITH OFFSET LAYERS**CROSS-REFERENCE TO RELATED APPLICATION**

This application is a Section 111(a) application relating to and claiming the benefit of commonly owned, U.S. Provisional Patent Application Ser. No. 63/401,976, filed Aug. 29, 2022, entitled "PHOTOVOLTAIC MODULES WITH OFFSET LAYERS," the contents of each of which are incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

The present invention relates to photovoltaic modules, and, more particularly, photovoltaic modules having offset layers.

BACKGROUND

Photovoltaic systems having solar panels are commonly installed on roofing of structures.

SUMMARY

In some embodiments, a system comprises a plurality of photovoltaic modules installed on a roof deck, wherein each of the photovoltaic modules includes at least one solar cell, a first encapsulant encapsulating the at least one solar cell, wherein the first encapsulant includes a first surface and a second surface opposite the first surface, a frontsheet juxtaposed with the first surface of the first encapsulant, wherein the frontsheet includes a glass layer, and a backsheet juxtaposed with the second surface of the first encapsulant, wherein the backsheet includes a first layer, wherein the first layer includes a first edge, a second edge opposite the first edge, a third edge extending from the first edge to the second edge, and a fourth edge opposite the third edge and extending from the first edge to the second edge, and a second layer adjacent the first layer, wherein the second layer includes a first edge, a second edge opposite the first edge of the second layer, a third edge extending from the first edge of the second layer to the second edge of the second layer, and a fourth edge opposite the third edge of the second layer and extending from the first edge of the second layer to the second edge of the second layer, wherein the first edge of the first layer is offset from the first edge of the second layer, wherein the second edge of the first layer is offset from the second edge of the second layer, wherein the third edge of the first layer is offset from the third edge of the second layer, and wherein the fourth edge of the first layer is offset from the fourth edge of the second layer.

In some embodiments, a second layer forms a lower surface of the photovoltaic module. In some embodiments, the first layer includes a first length extending from the first edge of the first layer to the second edge of the first layer, wherein the second layer includes a second length extending from the first edge of the second layer to the second edge of the second layer, and wherein the first length is greater than the second length. In some embodiments, the first layer includes a first width extending from the third edge of the first layer to the fourth edge of the first layer, wherein the second layer includes a second width extending from the third edge of the second layer to the fourth edge of the second layer, and wherein the first width is greater than the second width. In some embodiments, the second layer is

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composed of thermoplastic polyolefin (TPO). In some embodiments, the first layer is composed of a polymer. In some embodiments, the first layer is composed of thermoplastic polyolefin (TPO).

In some embodiments, the glass layer includes a first edge, a second edge opposite the first edge of the glass layer, a third edge extending from the first edge of the glass layer to the second edge of the glass layer, and a fourth edge opposite the third edge of the glass layer and extending from the first edge of the glass layer to the second edge of the glass layer, wherein the first edge of the glass layer is offset from the first edge of the second layer, wherein the second edge of the glass layer is offset from the second edge of the second layer, wherein the third edge of the glass layer is offset from the third edge of the second layer, and wherein the fourth edge of the glass layer is offset from the fourth edge of the second layer.

In some embodiments, the first layer includes a first length extending from the first edge of the first layer to the second edge of the first layer, wherein the second layer includes a second length extending from the first edge of the second layer to the second edge of the second layer, and wherein the second length is greater than the first length. In some embodiments, the first layer includes a first width extending from the third edge of the first layer to the fourth edge of the first layer, wherein the second layer includes a second width extending from the third edge of the second layer to the fourth edge of the second layer, and wherein the second width is greater than the first width.

In some embodiments, first layer is composed of thermoplastic polyolefin (TPO). In some embodiments, the second layer is composed of a polymer. In some embodiments, the second layer is composed of thermoplastic polyolefin (TPO).

In some embodiments, glass layer includes a first edge, a second edge opposite the first edge of the glass layer, a third edge extending from the first edge of the glass layer to the second edge of the glass layer, and a fourth edge opposite the third edge of the glass layer and extending from the first edge of the glass layer to the second edge of the glass layer, wherein the first edge of the glass layer is offset from the first edge of the first layer, wherein the second edge of the glass layer is offset from the second edge of the first layer, wherein the third edge of the glass layer is offset from the third edge of the first layer, and wherein the fourth edge of the glass layer is offset from the fourth edge of the first layer.

In some embodiments, the first edge of the first layer is offset from the first edge of the second layer by a first distance, wherein the second edge of the first layer is offset from the second edge of the second layer by a second distance, wherein the third edge of the first layer is offset from the third edge of the second layer by a third distance, and wherein the fourth edge of the first layer is offset from the fourth edge of the second layer by a fourth distance. In some embodiments, the first distance, the second distance, the third distance and the fourth distance are equal to one another. In some embodiments, the first distance is equal to the second distance, and wherein the third distance is equal to the fourth distance. In some embodiments, the first layer and the second layer are attached to one another by an adhesive layer. In some embodiments, the frontsheet includes a polymer layer and a second encapsulant, and wherein the polymer layer is attached to the glass layer by the second encapsulant.

In some embodiments, a photovoltaic module includes at least one solar cell, a first encapsulant encapsulating the at least one solar cell, wherein the first encapsulant includes a

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first surface and a second surface opposite the first surface, a frontsheet juxtaposed with the first surface of the first encapsulant, wherein the frontsheet includes a glass layer, and a backsheet juxtaposed with the second surface of the first encapsulant, wherein the backsheet includes a first layer, wherein the first layer includes a first edge, a second edge opposite the first edge, a third edge extending from the first edge to the second edge, and a fourth edge opposite the third edge and extending from the first edge to the second edge, and a second layer attached to the first layer, wherein the second layer includes a first edge, a second edge opposite the first edge of the second layer, a third edge extending from the first edge of the second layer to the second edge of the second layer, and a fourth edge opposite the third edge of the second layer and extending from the first edge of the second layer to the second edge of the second layer, wherein the first edge of the first layer is offset from the first edge of the second layer, wherein the second edge of the first layer is offset from the second edge of the second layer, wherein the third edge of the first layer is offset from the third edge of the second layer, and wherein the fourth edge of the first layer is offset from the fourth edge of the second layer, wherein the second layer forms a lower surface of the photovoltaic module, and wherein at least one of the first layer and the second layer is composed of thermoplastic polyolefin (TPO).

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic view of some embodiments of a photovoltaic module;

FIG. 2 is a perspective, schematic view of some embodiments of a photovoltaic module;

FIGS. 3 and 4 are schematic views of some embodiments of a photovoltaic module;

FIGS. 5 through 7 are schematic views of some embodiments of a photovoltaic module; and

FIGS. 8 through 10 show enlarged views of a portion of some embodiments of a photovoltaic module.

DETAILED DESCRIPTION

The present invention will be further explained with reference to the attached drawings, wherein like structures are referred to by like numerals throughout the several views. The drawings shown are not necessarily to scale, with emphasis instead generally being placed upon illustrating the principles of the present invention. Further, some features may be exaggerated to show details of particular components.

The figures constitute a part of this specification and include illustrative embodiments of the present invention and illustrate various objects and features thereof. Further, the figures are not necessarily to scale, some features may be exaggerated to show details of particular components. In addition, any measurements, specifications and the like shown in the figures are intended to be illustrative, and not restrictive. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present invention.

Among those benefits and improvements that have been disclosed, other objects and advantages of this invention will become apparent from the following description taken in conjunction with the accompanying figures. Detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodi-

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ments are merely illustrative of the invention that may be embodied in various forms. In addition, each of the examples given in connection with the various embodiments of the invention which are intended to be illustrative, and not restrictive.

Throughout the specification, the following terms take the meanings explicitly associated herein, unless the context clearly dictates otherwise. The phrases “in one embodiment” and “in some embodiments” as used herein do not necessarily refer to the same embodiment(s), though they may. Furthermore, the phrases “in another embodiment” and “in some other embodiments” as used herein do not necessarily refer to a different embodiment, although they may. Thus, as described below, various embodiments of the invention may be readily combined, without departing from the scope or spirit of the invention.

The term “based on” is not exclusive and allows for being based on additional factors not described, unless the context clearly dictates otherwise. In addition, throughout the specification, the meaning of “a,” “an,” and “the” include plural references. The meaning of “in” includes “in” and “on.”

Referring to FIGS. 1 through 3, in some embodiments, a photovoltaic module 10 includes at least one solar cell 12, an encapsulant 14 encapsulating the at least one solar cell 12, and a frontsheet 16 juxtaposed with the encapsulant 14. In some embodiments, the frontsheet 16 is juxtaposed with a first surface of the encapsulant 14. As used herein, the terms “encapsulating” and “encapsulates” mean to partially or fully envelope or enclose, and with respect to certain embodiments of the photovoltaic module 10, the at least one solar cell 12 is fully enveloped by or enclosed within the encapsulant 14, or partially enveloped by or enclosed within the encapsulant 14. In some embodiments, the encapsulant 14 encapsulates 50% to 99.9% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14 encapsulates 55% to 99.9% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14 encapsulates 60% to 99.9% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14 encapsulates 65% to 99.9% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14 encapsulates 70% to 99.9% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14 encapsulates 75% to 99.9% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14 encapsulates 80% to 99.9% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14 encapsulates 85% to 99.9% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14 encapsulates 90% to 99.9% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14 encapsulates 95% to 99.9% of an exterior surface area of the at least one solar cell 12.

In another embodiment, the encapsulant 14 encapsulates 50% to 95% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14 encapsulates 55% to 95% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14 encapsulates 60% to 95% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14 encapsulates 65% to 95% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14 encapsulates 70% to 95% of an exterior surface area of the at least one solar cell 12. In another embodiment, the encapsulant 14

In another embodiment, the encapsulant **14** encapsulates 50% to 75% of an exterior surface area of the at least one solar cell **12**. In another embodiment, the encapsulant **14** encapsulates 55% to 75% of an exterior surface area of the at least one solar cell **12**. In another embodiment, the encapsulant **14** encapsulates 60% to 75% of an exterior surface area of the at least one solar cell **12**. In another embodiment, the encapsulant **14** encapsulates 65% to 75% of an exterior surface area of the at least one solar cell **12**.

60 In some embodiments, the encapsulant **14** has a thickness of 1 mm to 4 mm. In another embodiment, the encapsulant **14** has a thickness of 1 mm to 3.5 mm. In another embodiment, the encapsulant **14** has a thickness of 1 mm to 3 mm. In another embodiment, the encapsulant **14** has a thickness of 1 mm to 2.5 mm. In another embodiment, the encapsulant **14** has a thickness of 1 mm to 2 mm. In another embodiment, the encapsulant **14** has a thickness of 1 mm to 1.5 mm. In

some embodiments, the encapsulant **14** has a thickness of 1.5 mm to 4 mm. In another embodiment, the encapsulant **14** has a thickness of 1.5 mm to 3.5 mm. In another embodiment, the encapsulant **14** has a thickness of 1.5 mm to 3 mm. In another embodiment, the encapsulant **14** has a thickness of 1.5 mm to 2.5 mm. In another embodiment, the encapsulant **14** has a thickness of 1.5 mm to 2 mm.

In some embodiments, the encapsulant **14** has a thickness of 2 mm to 4 mm. In another embodiment, the encapsulant **14** has a thickness of 2 mm to 3.5 mm. In another embodiment, the encapsulant **14** has a thickness of 2 mm to 3 mm. In another embodiment, the encapsulant **14** has a thickness of 2 mm to 2.5 mm. In some embodiments, the encapsulant **14** has a thickness of 2.5 mm to 4 mm. In another embodiment, the encapsulant **14** has a thickness of 2.5 mm to 3.5 mm. In another embodiment, the encapsulant **14** has a thickness of 2.5 mm to 3 mm. In some embodiments, the encapsulant **14** has a thickness of 3 mm to 4 mm. In another embodiment, the encapsulant **14** has a thickness of 3 mm to 3.5 mm. In some embodiments, the encapsulant **14** has a thickness of 3.5 mm to 4 mm.

In some embodiments, the encapsulant **14** has a thickness of 0.5 mm. In some embodiments, the encapsulant **14** has a thickness of 0.6 mm. In some embodiments, the encapsulant **14** has a thickness of 0.7 mm. In some embodiments, the encapsulant **14** has a thickness of 0.8 mm. In some embodiments, the encapsulant **14** has a thickness of 0.9 mm. In some embodiments, the encapsulant **14** has a thickness of 1 mm. In some embodiments, the encapsulant **14** has a thickness of 1.5 mm. In some embodiments, the encapsulant **14** has a thickness of 2 mm. In some embodiments, the encapsulant **14** has a thickness of 2.5 mm. In some embodiments, the encapsulant **14** has a thickness of 3 mm. In some embodiments, the encapsulant **14** has a thickness of 3.5 mm. In some embodiments, the encapsulant **14** has a thickness of 4 mm.

In some embodiments, a first layer **14a** of the encapsulant **14** has a thickness of 0.2 mm to 2 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 0.2 mm to 1.5 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 0.2 mm to 1 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 0.2 mm to 0.6 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 0.2 mm to 0.5 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 0.5 mm to 2 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 0.5 mm to 1.5 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 0.5 mm to 1 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 1 mm to 2 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 1 mm to 1.5 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 1.5 mm to 2 mm.

In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 0.2 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 0.3 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 0.4 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 0.45 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 0.5 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 0.6 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 1 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has

a thickness of 1.5 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 2 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 2.5 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 3 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 3.5 mm. In some embodiments, the first layer **14a** of the encapsulant **14** has a thickness of 4 mm.

In some embodiments, a second layer **14b** of the encapsulant **14** has a thickness of 0.2 mm to 2 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 0.2 mm to 1.5 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 0.2 mm to 1 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 0.2 mm to 0.5 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 0.5 mm to 2 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 0.5 mm to 1.5 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 0.5 mm to 1 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 1 mm to 2 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 1 mm to 1.5 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 1.5 mm to 2 mm.

In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 0.2 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 0.3 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 0.4 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 0.45 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 0.5 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 1 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 1.5 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 2 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 2.5 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 3 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 3.5 mm. In some embodiments, the second layer **14b** of the encapsulant **14** has a thickness of 4 mm.

In some embodiments, the thickness of the first layer **14a** is equal to the thickness of the second layer **14b**. In another embodiment, the thickness of the first layer **14a** is different from the thickness of the second layer **14b**.

In some embodiments, the encapsulant **14** may be made from polyolefins, ethyl vinyl acetates, ionomers, silicones, poly vinyl butyral, epoxies, polyurethanes, or combinations/hybrids thereof. In some embodiments, the encapsulant **14** is made from thermosetting polyolefin.

In some embodiments, the photovoltaic module **10** includes a first end **18**, a second end **20** opposite the first end **18**, a first surface **19** extending from the first end **18** to the second end **20**, and a second surface **21** opposite the first surface **19** and extending from the first end **18** to the second end **20**. In some embodiments, the first surface **19** is an upper, sun facing-side surface of the photovoltaic module **10**, and the second surface **21** is a lower surface configured to face a roof deck on which the photovoltaic module **10** is installed on the roof deck. In some embodiments, the polymer layer **24** includes the first surface **19**. In some embodiments, the first surface **19** is textured. In some

embodiments, the first surface **19** is textured to impart an appearance of a traditional asphalt roofing shingle. In some embodiments, the first surface **19** is textured to impart an appearance of and aesthetically match a roofing shingle. In some embodiments, the first surface **19** is an embossed surface. In some embodiments, a pattern on the first surface **19** includes a plurality of indentations. In some embodiment, each of the plurality of indentations has a circular shape. In some embodiment, each of the plurality of indentations has a rectangular shape. In some embodiments, each of the plurality of indentations has a square shape. In some embodiments, each of the plurality of indentations has a triangular shape. In some embodiments, each of the plurality of indentations has an elliptical shape. In some embodiments, each of the plurality of indentations has an oval shape. In some embodiments, each of the plurality of indentations has a rhombus shape. In some embodiments, each of the plurality of indentations has a hexagonal shape. In some embodiments, each of the plurality of indentations includes a pentagonal shape. In some embodiments, each of the plurality of indentations has a polygonal shape. In some embodiments, each of the plurality of indentations has a non-polygonal shape. In some embodiments, each of the plurality of indentations has a geometric shape. In some embodiments, each of the plurality of indentations has a non-geometric shape. In some embodiments, the pattern on the first surface **19** includes a plurality of dimples. In some embodiments, the indentations are created by embossing a portion of the first surface **19**. In some embodiments, the texture includes a surface roughness (Ra). In some embodiments, the surface roughness (Ra) is 1 micron to 200 microns. In some embodiments, the pattern on the first surface **19** includes a plurality of lines. In some embodiments, the pattern on the first surface **19** includes a plurality of grooves. In some embodiments, the pattern on the first surface **19** includes crosshatches.

In some embodiments, the at least one solar cell **12** includes a plurality of the solar cells **12**. In some embodiments, the plurality of solar cells **12** includes two solar cells. In some embodiments, the plurality of solar cells **12** includes three solar cells. In some embodiments, the plurality of solar cells **12** includes four solar cells. In some embodiments, the plurality of solar cells **12** includes five solar cells. In some embodiments, the plurality of solar cells **12** includes six solar cells. In some embodiments, the plurality of solar cells **12** includes seven solar cells. In some embodiments, the plurality of solar cells **12** includes eight solar cells. In some embodiments, the plurality of solar cells **12** includes nine solar cells. In some embodiments, the plurality of solar cells **12** includes ten solar cells. In some embodiments, the plurality of solar cells **12** includes eleven solar cells. In some embodiments, the plurality of solar cells **12** includes twelve solar cells. In some embodiments, the plurality of solar cells **12** includes thirteen solar cells. In some embodiments, the plurality of solar cells **12** includes fourteen solar cells. In some embodiments, the plurality of solar cells **12** includes fifteen solar cells. In some embodiments, the plurality of solar cells **12** includes sixteen solar cells. In some embodiments, the plurality of solar cells **12** includes more than sixteen solar cells.

In some embodiments, the plurality of solar cells **12** is arranged in one row (i.e., one reveal). In another embodiment, the plurality of solar cells **12** is arranged in two rows (i.e., two reveals). In another embodiment, the plurality of solar cells **12** is arranged in three rows (i.e., three reveals). In another embodiment, the plurality of solar cells **12** is arranged in four rows (i.e., four reveals). In another embodi-

ment, the plurality of solar cells **12** is arranged in five rows (i.e., five reveals). In another embodiment, the plurality of solar cells **12** is arranged in six rows (i.e., six reveals). In other embodiments, the plurality of solar cells **12** is arranged in more than six rows. In some embodiments, the at least one solar cell **12** is electrically inactive (i.e., a “dummy” solar cell). In some embodiments, the photovoltaic module **10** does not include the at least one solar cell **12**.

In some embodiments, the frontsheet **16** includes a glass layer **22** and a polymer layer **24** attached to a first surface of the glass layer **22**. In some embodiments, the frontsheet **16** is juxtaposed with the first layer **14a** of the encapsulant **14**. In some embodiments, each of the encapsulant **14**, the glass layer **22**, and the polymer layer **24** is transparent. In some embodiments, the polymer layer **24** is attached to the glass layer **22** by an first adhesive layer **26**. In some embodiments, the first adhesive layer **26** may include polyvinyl butyrate, acrylic, silicone, or polycarbonate. In another embodiment, the first adhesive layer **26** may include pressure sensitive adhesives. In another embodiment, the polymer layer **24** is attached to the glass layer **22** by thermal bonding. In another embodiment, the frontsheet **16** includes at least one of the glass layer **22** or the polymer layer **24**. In some embodiments, the first adhesive layer **26** is transparent. As used herein, the term “transparent” means having a solar weighted transmittance of 80% or greater, and with respect to certain embodiments of the photovoltaic module **10**, a transparent layer of the photovoltaic module has a solar weighted transmittance of 80% or greater. In another embodiment, the frontsheet **16** does not include the glass layer **22**.

In some embodiments, the glass layer **22** has a thickness of 1 mm to 4 mm. In some embodiments, the glass layer **22** has a thickness of 1 mm to 3.5 mm. In some embodiments, the glass layer **22** has a thickness of 1 mm to 3 mm. In some embodiments, the glass layer **22** has a thickness of 1 mm to 2.5 mm. In some embodiments, the glass layer **22** has a thickness of 1 mm to 2 mm. In some embodiments, the glass layer **22** has a thickness of 1 mm to 1.5 mm. In some embodiments, the glass layer **22** has a thickness of 1.5 mm to 4 mm. In some embodiments, the glass layer **22** has a thickness of 1.5 mm to 3.5 mm. In some embodiments, the glass layer **22** has a thickness of 1.5 mm to 3 mm. In some embodiments, the glass layer **22** has a thickness of 1 mm to 2.5 mm. In some embodiments, the glass layer **22** has a thickness of 1.5 mm to 2 mm.

In some embodiments, the glass layer **22** has a thickness of 2 mm to 4 mm. In some embodiments, the glass layer **22** has a thickness of 2 mm to 3.5 mm. In some embodiments, the glass layer **22** has a thickness of 2 mm to 3 mm. In some embodiments, the glass layer **22** has a thickness of 2 mm to 2.5 mm. In some embodiments, the glass layer **22** has a thickness of 2.5 mm to 4 mm. In some embodiments, the glass layer **22** has a thickness of 2.5 mm to 3.5 mm. In some embodiments, the glass layer **22** has a thickness of 2.5 mm to 3 mm. In some embodiments, the glass layer **22** has a thickness of 3 mm to 4 mm. In some embodiments, the glass layer **22** has a thickness of 3 mm to 3.5 mm. In some embodiments, the glass layer **22** has a thickness of 3.5 mm to 4 mm.

In some embodiments, the glass layer **22** has a thickness of 1 mm. In some embodiments, the glass layer **22** has a thickness of 1.5 mm. In some embodiments, the glass layer **22** has a thickness of 2 mm. In some embodiments, the glass layer **22** has a thickness of 2.5 mm. In some embodiments, the glass layer **22** has a thickness of 3 mm. In some

In some embodiments, the first adhesive layer **26** has a thickness of 300 μm to 900 μm . In some embodiments, the first adhesive layer **26** has a thickness of 300 μm to 850 μm . In some embodiments, the first adhesive layer **26** has a thickness of 300 μm to 800 μm . In some embodiments, the first adhesive layer **26** has a thickness of 300 μm to 750 μm . In some embodiments, the first adhesive layer **26** has a thickness of 300 μm to 700 μm . In some embodiments, the first adhesive layer **26** has a thickness of 300 μm to 650 μm . In some embodiments, the first adhesive layer **26** has a thickness of 300 μm to 600 μm . In some embodiments, the first adhesive layer **26** has a thickness of 300 μm to 550 μm . In some embodiments, the first adhesive layer **26** has a thickness of 300 μm to 500 μm . In some embodiments, the first adhesive layer **26** has a thickness of 300 μm to 450 μm . In some embodiments, the first adhesive layer **26** has a

In some embodiments, the first adhesive layer 26 has a thickness of 550 μm to 900 μm . In some embodiments, the first adhesive layer 26 has a thickness of 550 μm to 850 μm . In some embodiments, the first adhesive layer 26 has a thickness of 550 μm to 800 μm . In some embodiments, the first adhesive layer 26 has a thickness of 550 μm to 750 μm . In some embodiments, the first adhesive layer 26 has a

In some embodiments, the polymer layer **24** is composed of a fluoropolymer. In certain embodiments, the fluoropolymer may be ethylene tetrafluoroethylene (ETFE), fluoropolymer is polyvinylidene fluoride (PVDF), tetrafluoro-

In some embodiments, the polymer layer **24** has a thickness of 0.1 mm to 0.5 mm. In some embodiments, the polymer layer **24** has a thickness of 0.1 mm to 0.45 mm. In some embodiments, the polymer layer **24** has a thickness of 0.1 mm to 0.4 mm. In some embodiments, the polymer layer **24** has a thickness of 0.1 mm to 0.35 mm. In some embodiments, the polymer layer **24** has a thickness of 0.1 mm to 0.3 mm. In some embodiments, the polymer layer **24** has a thickness of 0.1 mm to 0.25 mm. In some embodiments, the polymer layer **24** has a thickness of 0.1 mm to 0.2 mm. In some embodiments, the polymer layer **24** has a thickness of 0.1 mm to 0.15 mm. In some embodiments, the polymer layer **24** has a thickness of 0.15 mm to 0.5 mm. In some embodiments, the polymer layer **24** has a thickness of 0.15 mm to 0.45 mm. In some embodiments, the polymer layer **24** has a thickness of 0.15 mm to 0.4 mm. In some embodiments, the polymer layer **24** has a thickness of 0.15 mm to 0.35 mm. In some embodiments, the polymer layer **24** has a thickness of 0.15 mm to 0.3 mm. In some embodiments, the polymer layer **24** has a thickness of 0.15 mm to 0.25 mm. In some embodiments, the polymer layer **24** has a thickness of 0.15 mm to 0.2 mm.

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In some embodiments, the polymer layer **24** has a thickness of 0.2 mm to 0.5 mm. In some embodiments, the polymer layer **24** has a thickness of 0.2 mm to 0.45 mm. In some embodiments, the polymer layer **24** has a thickness of 0.2 mm to 0.4 mm. In some embodiments, the polymer layer **24** has a thickness of 0.2 mm to 0.35 mm. In some embodiments, the polymer layer **24** has a thickness of 0.2 mm to 0.3 mm. In some embodiments, the polymer layer **24** has a thickness of 0.2 mm to 0.25 mm. In some embodiments, the polymer layer **24** has a thickness of 0.25 mm to 0.5 mm. In some embodiments, the polymer layer **24** has a thickness of 0.25 mm to 0.45 mm. In some embodiments, the polymer layer **24** has a thickness of 0.25 mm to 0.4 mm. In some embodiments, the polymer layer **24** has a thickness of 0.25 mm to 0.35 mm. In some embodiments, the polymer layer **24** has a thickness of 0.25 mm to 0.3 mm.

In some embodiments, the polymer layer **24** has a thickness of 0.3 mm to 0.5 mm. In some embodiments, the polymer layer **24** has a thickness of 0.3 mm to 0.45 mm. In some embodiments, the polymer layer **24** has a thickness of 0.3 mm to 0.4 mm. In some embodiments, the polymer layer **24** has a thickness of 0.3 mm to 0.35 mm. In some embodiments, the polymer layer **24** has a thickness of 0.35 mm to 0.5 mm. In some embodiments, the polymer layer **24** has a thickness of 0.35 mm to 0.45 mm. In some embodiments, the polymer layer **24** has a thickness of 0.35 mm to 0.4 mm. In some embodiments, the polymer layer **24** has a thickness of 0.4 mm to 0.5 mm. In some embodiments, the polymer layer **24** has a thickness of 0.4 mm to 0.45 mm. In some embodiments, the polymer layer **24** has a thickness of 0.45 mm to 0.5 mm.

In some embodiments, the polymer layer **24** has a thickness of 0.01 mm. In some embodiments, the polymer layer **24** has a thickness of 0.05 mm. In some embodiments, the polymer layer **24** has a thickness of 0.1 mm. In some embodiments, the polymer layer **24** has a thickness of 0.15 mm. In some embodiments, the polymer layer **24** has a thickness of 0.2 mm. In some embodiments, the polymer layer **24** has a thickness of 0.25 mm. In some embodiments, the polymer layer **24** has a thickness of 0.3 mm. In some embodiments, the polymer layer **24** has a thickness of 0.35 mm. In some embodiments, the polymer layer **24** has a thickness of 0.4 mm. In some embodiments, the polymer layer **24** has a thickness of 0.45 mm. In some embodiments, the polymer layer **24** has a thickness of 0.5 mm.

In some embodiments, a backsheet **28** is juxtaposed with a second layer **14b** of the encapsulant **14**. In some embodiments, the backsheet **28** includes a first layer **30**. In some embodiments, the backsheet **28** includes a second layer **32**. In some embodiments, the first layer **30** is juxtaposed with the second layer **14b** of the encapsulant **14**. In some embodiments, the second layer **32** is juxtaposed with the first layer **30**. In some embodiments, the backsheet **28** only includes the first layer **30** (see FIG. 3). In some embodiments, the backsheet **28** includes a scrim. In some embodiments, the scrim is located between the first layer **30** and the second layer **32**.

In some embodiments, the first layer **30** of the backsheet **28** is composed of a polymeric material. In some embodiments, the first layer **30** of the backsheet **28** is composed of polyethylene terephthalate ("PET"). In another embodiment, the first layer **30** of the backsheet **28** is composed of ethylene tetrafluoroethylene ("ETFE"). In some embodiments, the first layer **30** of the backsheet **28** is composed of an acrylic such as polymethyl methacrylate ("PMMA"). In some embodiments, the first layer **30** of the backsheet **28** is composed of thermoplastic polyolefin (TPO). In some

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embodiments, the first layer **30** of the backsheet **28** includes of a single ply TPO roofing membrane. In other embodiments, non-limiting examples of TPO membranes are disclosed in U.S. Pat. No. 9,359,014 to Yang et al., which is incorporated by reference herein in its entirety. In another embodiment, the first layer **30** of the backsheet **28** is composed of polyvinyl chloride. In some embodiments, the first layer **30** of the backsheet **28** is composed of ethylene propylene diene monomer (EPDM) rubber. In some embodiments, the first layer **30** of the backsheet **28** includes a flame retardant additive. In some embodiments, the flame retardant additive may be clays, nanoclays, silicas, carbon black, metal hydroxides such as aluminum hydroxide, metal foils, graphite, and combinations thereof. In some embodiments, the first layer **30** is white in color. In some embodiments, the first layer **30** is white TPO.

In some embodiments, the first layer **30** has a thickness of 0.2 mm to 0.5 mm. In another embodiment, the first layer **30** has a thickness of 0.2 mm to 0.4 mm. In another embodiment, the first layer **30** has a thickness of 0.2 mm to 0.3 mm. In some embodiments, the first layer **30** has a thickness of 0.3 mm to 0.5 mm. In another embodiment, the first layer **30** has a thickness of 0.3 mm to 0.4 mm. In some embodiments, the first layer **30** has a thickness of 0.4 mm to 0.5 mm. In some embodiments, the first layer **30** has a thickness of 0.2 mm. In some embodiments, the first layer **30** has a thickness of 0.3 mm. In some embodiments, the first layer **30** has a thickness of 0.4 mm. In some embodiments, the first layer **30** has a thickness of 0.5 mm.

In some embodiments, the second layer **32** of the backsheet **28** is composed of a polymeric material. In some embodiments, the second layer **32** of the backsheet **28** is composed of thermoplastic polyolefin (TPO). In some embodiments, the second layer **32** of the backsheet **28** includes a single ply TPO roofing membrane. In other embodiments, non-limiting examples of TPO membranes are disclosed in U.S. Pat. No. 9,359,014 to Yang et al., which is incorporated by reference herein in its entirety. In another embodiment, the second layer **32** of the backsheet **28** is composed of polyethylene terephthalate ("PET"). In another embodiment, the second layer **32** is composed of styrene acrylic copolymer. In another embodiment, the second layer **32** of the backsheet **28** is composed of ethylene tetrafluoroethylene ("ETFE"). In some embodiments, the second layer **32** of the backsheet **28** is composed of an acrylic such as polymethyl methacrylate ("PMMA"). In another embodiment, the second layer **32** of the backsheet **28** is composed of polyvinyl chloride. In some embodiments, the second layer **32** of the backsheet **28** is composed of ethylene propylene diene monomer (EPDM) rubber. In some embodiments, the second layer **32** of the backsheet **28** includes a flame retardant additive. In some embodiments, the flame retardant additive may be clays, nanoclays, silicas, carbon black, metal hydroxides such as aluminum hydroxide, metal foils, graphite, and combinations thereof. In some embodiments, the second layer **32** is white in color. In some embodiments, the second layer **32** is white TPO.

In some embodiments, the second layer **32** has a thickness of 1 mm to 5 mm. In some embodiments, the second layer **32** has a thickness of 1 mm to 4 mm. In some embodiments, the second layer **32** has a thickness of 1 mm to 3 mm. In some embodiments, the second layer **32** has a thickness of 1 mm to 2 mm. In some embodiments, the second layer **32** has a thickness of 2 mm to 5 mm. In some embodiments, the second layer **32** has a thickness of 2 mm to 4 mm. In some embodiments, the second layer **32** has a thickness of 2 mm to 3 mm. In some embodiments, the second layer **32** has a

In some embodiments, the second adhesive layer **34** has a thickness of 300 μm to 900 μm . In some embodiments, the

In some embodiments, the second adhesive layer **34** has a thickness of 500 μm to 900 μm . In some embodiments, the second adhesive layer **34** has a thickness of 500 μm to 850

In some embodiments, at least one of the first, second, third and fourth edges **30a**, **30b**, **30c**, **30d** of the first layer **30** is offset from the corresponding first, second, third and fourth edges **32a**, **32b**, **32c**, **32d** of the second layer **32**, while the remaining ones of the first, second, third and fourth

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edges **30a**, **30b**, **30c**, **30d** of the first layer **30** are not offset from the corresponding first, second, third and fourth edges **32a**, **32b**, **32c**, **32d** of the second layer **32**. In some embodiments, at least two of the first, second, third and fourth edges **30a**, **30b**, **30c**, **30d** of the first layer **30** is offset from the corresponding first, second, third and fourth edges **32a**, **32b**, **32c**, **32d** of the second layer **32**, while the remaining ones of the first, second, third and fourth edges **30a**, **30b**, **30c**, **30d** of the first layer **30** are not offset from the corresponding first, second, third and fourth edges **32a**, **32b**, **32c**, **32d** of the second layer **32**. In some embodiments, at least three of the first, second, third and fourth edges **30a**, **30b**, **30c**, **30d** of the first layer **30** is offset from the corresponding first, second, third and fourth edges **32a**, **32b**, **32c**, **32d** of the second layer **32**, while the remaining ones of the first, second, third and fourth edges **30a**, **30b**, **30c**, **30d** of the first layer **30** are not offset from the corresponding first, second, third and fourth edges **32a**, **32b**, **32c**, **32d** of the second layer **32**.

In some embodiments, the second layer **32** forms a lower surface of the photovoltaic module **10**. In some embodiments, the first layer **30** forms a lower surface of the photovoltaic module **10** (see FIG. 4).

In some embodiments, the first layer **30** includes a first length **L1** extending from the first edge **30a** of the first layer **30** to the second edge of the first layer **30**. In some embodiments, the second layer **32** includes a second length **L2** extending from the first edge **32a** of the second layer **32** to the second edge **32b** of the second layer **32**. Referring to FIG. 1, in some embodiments, the first length **L1** is greater than the second length **L2**. Referring to FIG. 3, in some embodiments, the second length **L2** is greater than the first length **L1**.

In some embodiments, the first layer **30** includes a first width **W1** extending from the third edge **30c** of the first layer **30** to the fourth edge **30d** of the first layer **30**. In some embodiments, the second layer **32** includes a second width **W2** extending from the third edge **32c** of the second layer **32** to the fourth edge **32d** of the second layer **32**. In some embodiments, the first width **W1** is greater than the second width **W2**. In some embodiments, the second width **W2** is greater than the first width **W1**.

Referring to FIG. 5, in some embodiments, the first edge **30a** of the first layer **30** is offset from the first edge **32a** of the second layer **32** by a first distance **D1**. In some embodiments, the second edge **30b** of the first layer **30** is offset from the second edge **32b** of the second layer by a second distance **D2**. In some embodiments, the third edge **30c** of the first layer **30** is offset from the third edge **32c** of the second layer **32** by a third distance **D3**. In some embodiments, the fourth edge **30d** of the first layer **30** is offset from the fourth edge **32d** of the second layer **32** by a fourth distance **D4**.

In some embodiments, the first distance **D1**, the second distance **D2**, the third distance **D3** and the fourth distance **D4** are equal to one another. In some embodiments, the first distance **D1**, the second distance **D2**, the third distance **D3** and the fourth distance **D4** are all different from one another. In some embodiments, the first distance **D1** is equal to the second distance **D2**. In some embodiments, the third distance **D3** is equal to the fourth distance **D4**.

In some embodiments, the first distance **D1** is 1 mm to 10 mm. In some embodiments, the first distance **D1** is 2 mm to 10 mm. In some embodiments, the first distance **D1** is 3 mm to 10 mm. In some embodiments, the first distance **D1** is 4 mm to 10 mm. In some embodiments, the first distance **D1** is 5 mm to 10 mm. In some embodiments, the first distance **D1** is 6 mm to 10 mm. In some embodiments, the first distance **D1** is 7 mm to 10 mm. In some embodiments, the

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first distance **D1** is 8 mm to 10 mm. In some embodiments, the first distance **D1** is 9 mm to 10 mm. In some embodiments, the first distance **D1** is 1 mm. In some embodiments, the first distance **D1** is 2 mm. In some embodiments, the first distance **D1** is 3 mm. In some embodiments, the first distance **D1** is 4 mm. In some embodiments, the first distance **D1** is 5 mm. In some embodiments, the first distance **D1** is 6 mm. In some embodiments, the first distance **D1** is 7 mm. In some embodiments, the first distance **D1** is 8 mm. In some embodiments, the first distance **D1** is 9 mm. In some embodiments, the first distance **D1** is 10 mm.

In some embodiments, the first distance **D2** is 1 mm to 10 mm. In some embodiments, the first distance **D2** is 2 mm to 10 mm. In some embodiments, the first distance **D2** is 3 mm to 10 mm. In some embodiments, the first distance **D2** is 4 mm to 10 mm. In some embodiments, the first distance **D2** is 5 mm to 10 mm. In some embodiments, the first distance **D2** is 6 mm to 10 mm. In some embodiments, the first distance **D2** is 7 mm to 10 mm. In some embodiments, the first distance **D2** is 8 mm to 10 mm. In some embodiments, the first distance **D2** is 9 mm to 10 mm. In some embodiments, the first distance **D2** is 1 mm. In some embodiments, the first distance **D2** is 2 mm. In some embodiments, the first distance **D2** is 3 mm. In some embodiments, the first distance **D2** is 4 mm. In some embodiments, the first distance **D2** is 5 mm. In some embodiments, the first distance **D2** is 6 mm. In some embodiments, the first distance **D2** is 7 mm. In some embodiments, the first distance **D2** is 8 mm. In some embodiments, the first distance **D2** is 9 mm. In some embodiments, the first distance **D2** is 10 mm.

In some embodiments, the first distance **D3** is 1 mm to 10 mm. In some embodiments, the first distance **D3** is 2 mm to 10 mm. In some embodiments, the first distance **D3** is 3 mm to 10 mm. In some embodiments, the first distance **D3** is 4 mm to 10 mm. In some embodiments, the first distance **D3** is 5 mm to 10 mm. In some embodiments, the first distance **D3** is 6 mm to 10 mm. In some embodiments, the first distance **D3** is 7 mm to 10 mm. In some embodiments, the first distance **D3** is 8 mm to 10 mm. In some embodiments, the first distance **D3** is 9 mm to 10 mm. In some embodiments, the first distance **D3** is 1 mm. In some embodiments, the first distance **D3** is 2 mm. In some embodiments, the first distance **D3** is 3 mm. In some embodiments, the first distance **D3** is 4 mm. In some embodiments, the first distance **D3** is 5 mm. In some embodiments, the first distance **D3** is 6 mm. In some embodiments, the first distance **D3** is 7 mm. In some embodiments, the first distance **D3** is 8 mm. In some embodiments, the first distance **D3** is 9 mm. In some embodiments, the first distance **D3** is 10 mm.

In some embodiments, the first distance **D4** is 1 mm to 10 mm. In some embodiments, the first distance **D4** is 2 mm to 10 mm. In some embodiments, the first distance **D4** is 3 mm to 10 mm. In some embodiments, the first distance **D4** is 4 mm to 10 mm. In some embodiments, the first distance **D4** is 5 mm to 10 mm. In some embodiments, the first distance **D4** is 6 mm to 10 mm. In some embodiments, the first distance **D4** is 7 mm to 10 mm. In some embodiments, the first distance **D4** is 8 mm to 10 mm. In some embodiments, the first distance **D4** is 9 mm to 10 mm. In some embodiments, the first distance **D4** is 1 mm. In some embodiments, the first distance **D4** is 2 mm. In some embodiments, the first distance **D4** is 3 mm. In some embodiments, the first distance **D4** is 4 mm. In some embodiments, the first distance **D4** is 5 mm. In some embodiments, the first

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distance D4 is 6 mm. In some embodiments, the first distance D4 is 7 mm. In some embodiments, the first distance D4 is 8 mm. In some embodiments, the first distance D4 is 9 mm. In some embodiments, the first distance D4 is 10 mm.

Referring to FIGS. 1 through 4, 6 and 7, In some embodiments, the glass layer 22 includes a first edge 22a, a second edge 22b opposite the first edge 22a, a third edge 22c extending from the first edge 22a to the second edge 22b, and a fourth edge 22d opposite the third edge 22c and extending from the first edge 22a to the second edge 22b. In some embodiments, the first edge 22a is offset from the first edge 32a of the second layer 32.

In some embodiments, the first edge 22a is offset from the first edge 32a of the second layer 32 by the first distance D1. In some embodiments, the second edge 22b of the glass layer 22 is offset from the second edge 32b of the second layer 32. In some embodiments, the second edge 22b of the glass layer 22 is offset from the second edge 32b of the second layer 32 by the second distance D2. In some embodiments, the third edge 22c of the glass layer 22 is offset from the third edge 32c of the second layer 32. In some embodiments, the third edge 22c of the glass layer 22 is offset from the third edge 32c of the second layer 32 by the third distance D3. In some embodiments, the fourth edge 22d of the glass layer 22 is offset from the fourth edge 32d of the second layer 32. In some embodiments, the fourth edge 22d of the glass layer 22 is offset from the fourth edge 32d of the second layer 32 by the fourth distance D4.

In some embodiments, the first edge 22a of the glass layer 22 is offset from the first edge 30a of the first layer 30. In some embodiments, the first edge 22a of the glass layer 22 is offset from the first edge 30a of the first layer 30 by the first distance D1. In some embodiments, the second edge 22b of the glass layer 22 is offset from the second edge 30b of the first layer 30. In some embodiments, the second edge 22b of the glass layer 22 is offset from the second edge 30b of the first layer 30 by the second distance D2. In some embodiments, the third edge 22c of the glass layer 22 is offset from the third edge 30c of the first layer 30. In some embodiments, the third edge 22c of the glass layer 22 is offset from the third edge 30c of the first layer 30 by the third distance D3. In some embodiments, the fourth edge 22d of the glass layer 22 is offset from the fourth edge 30d of the first layer 30. In some embodiments, the fourth edge 22d of the glass layer 22 is offset from the fourth edge 30d of the first layer 30 by the fourth distance D4.

In some embodiments, the photovoltaic module 10 is laminated. In some embodiments, all of the previously mentioned applicable layers of the photovoltaic module 10 are laminated in one lamination step. In some embodiments, a method of making the photovoltaic module 10 includes a first lamination step and a second lamination step. In an embodiment, the first lamination step includes laminating the layers of the frontsheet 16. In an embodiment, the first lamination step includes laminating the glass layer 22, the polymer layer 22, and the adhesive layer 24. In some embodiments, the second lamination step includes laminating the frontsheet 16 with the remaining layers of the photovoltaic module 10. In an embodiment, the second lamination step includes laminating the frontsheet 16, the encapsulant 14, the at least one solar cell 12, and the backsheet 28.

In some embodiments, a headlap portion is attached to the backsheet 28. In some embodiments, the headlap portion is attached to the first layer 30. In some embodiments, the headlap portion is attached to the second layer 32. In some

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embodiments, headlap portion and the backsheet 28 are welded to one another. In some embodiments, headlap portion and the backsheet 28 are ultrasonically welded to one another. In some embodiments, headlap portion and the backsheet 28 are heat welded to one another. In some embodiments, headlap portion and the backsheet 28 are thermally bonded to one another. In some embodiments, headlap portion and the backsheet 28 are adhered to one another by an adhesive. In some embodiments, headlap portion and the backsheet 28 are laminated. In some embodiments, headlap portion and the backsheet 28 are co-extruded. In some embodiments, headlap portion and the backsheet 28 are mechanically attached to one another. In some embodiments, headlap portion and the backsheet 28 are attached to one another by at least one fastener. In some embodiments, headlap portion and the backsheet 28 are attached to one another by a plurality of fasteners. In some embodiments, headlap portion and the backsheet 28 are attached to one another by deforming one of headlap portion and the backsheet 28 into the other one of headlap portion and the backsheet 28. In some embodiments, headlap portion and the backsheet 28 are attached to one another by deforming a portion of one of headlap portion and the backsheet 28 into the other one of headlap portion and the backsheet 28. In some embodiments, headlap portion and the backsheet 28 are attached to one another by forming at least one hole in one or both of headlap portion and the backsheet 28 and dispensing molten material into the at least one hole to connect headlap portion and the backsheet 28. In some embodiments, non-limiting examples of fasteners, fastening means and methods for fastening, connecting and attaching the first layer 30 to the second layer 32 are disclosed in U.S. Pat. No. 7,833,371 to Binkley et al, U.S. Pat. No. 8,006,457 to Binkley et al, U.S. Pat. No. 8,127,514 to Binkley et al, and U.S. Pat. No. 8,316,608 to Binkley et al, each of which is incorporated by reference herein in its entirety.

Referring to FIGS. 8 through 10, in some embodiments, the offset of the backsheet 28 (i.e., either the first layer 30 or the second layer 32) from the glass layer 22 reduces pinchoff on the edges of the photovoltaic module 10. In some embodiments, as used herein, the terms "pinchoff" or "pinch" is that the thicknesses of the layers of the photovoltaic module 10 causes pressure at the edges 22a through 22d of the glass layer 22, which can squeeze the encapsulant 14 while in a liquid form during lamination of the photovoltaic module 10, and could result in reduced bond line thickness. In some embodiments, pinchoff can result in premature delamination of the backsheet 28 from the glass layer 22. In some embodiments, the offset of the backsheet 28 (i.e., either the first layer 30 or the second layer 32) from the glass layer 22 reduces visibility of any scrim included within the backsheet 28. In some embodiments, the offset of one or more of the edges 32a, 32b, 32c, 32d of the second layer 32 relative to the corresponding edges 30a, 30b, 30c, 30d of the first layer 30 reduces the potential of the first layer 30 to pinch. In some embodiments, the offset of one or more of the edges 30a, 30b, 30c, 30d of the first layer 30 relative to the corresponding edges 32a, 32b, 32c, 32d of the second layer 32 reduces the potential of the second layer 32 to pinch. In some embodiments, each of the distances D1 through D4 is of a sufficient amount so as to enable the edges 30a through 30d of the first layer 30 and/or the edges 32a through 32d of the second layer 32 to reduce the potential of the first layer 30 and the second layer 32 to pinch.

In some embodiments, each of the distances D1 through D4 is of a sufficient amount so as to enable the edges 30a

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through 30d of the first layer 30 and/or the edges 32a through 32d of the second layer 32, as applicable, to be cut or trimmed when the photovoltaic module 10 is installed on a roof deck 90 for aesthetically matching the photovoltaic module 10 with other ones of the photovoltaic module 10.

In some embodiments, the roof deck 90 is a steep slope roof deck. As defined herein, a “steep slope roof deck” is any roof deck that is disposed on a roof having a pitch of Y/X, where Y and X are in a ratio of 4:12 to 12:12, where Y corresponds to the “rise” of the roof, and where X corresponds to the “run” of the roof.

In some embodiments, the roof deck 90 may be a sloped roof of a structure. As used herein, a “sloped” roof deck is a roof deck that has a slope less than a slope of a steep slope roof deck, but is not a flat roof deck.

In some embodiments, the roof deck 90 is a component of a commercial roof. In some embodiments, the roof deck 90 has a slope of 3 inches per foot or less. In some embodiments, the roof deck 90 has a slope of 0.25 inch to 3 inches per foot. In some embodiments, the roof deck 90 has a slope of 0.25 inch to 2 inches per foot. In some embodiments, the roof deck 90 has a slope of 0.25 inch to 1 inch per foot. In some embodiments, the roof deck 90 has a slope of 1 inch to 3 inches per foot. In some embodiments, the roof deck 90 has a slope of 1 inch to 2 inches per foot. In some embodiments, the roof deck 90 has a slope of 2 inches to 3 inches per foot.

In some embodiments, a system of at least one of the photovoltaic modules 10 is installed on the roof deck 90. In some embodiments, a system of a plurality of the photovoltaic modules 10 is installed on the roof deck 90. In some embodiments, the plurality of the photovoltaic modules 10 includes two of the photovoltaic modules 10. In some embodiments, the plurality of the photovoltaic modules 10 includes three of the photovoltaic modules 10. In some embodiments, the plurality of the photovoltaic modules 10 includes four of the photovoltaic modules 10. In some embodiments, the plurality of the photovoltaic modules 10 includes five of the photovoltaic modules 10. In some embodiments, the plurality of the photovoltaic modules 10 includes six of the photovoltaic modules 10. In some embodiments, the plurality of the photovoltaic modules 10 includes more than six of the photovoltaic modules 10. In some embodiments, the plurality of the photovoltaic modules 10 is installed in an array on the roof deck 90. In some embodiments, the array includes a plurality of subarrays. In some embodiments, the array includes two subarrays. In some embodiments, the array includes three subarrays. In some embodiments, the array includes four subarrays. In some embodiments, the array includes five subarrays. In some embodiments, the array includes six subarrays. In some embodiments, the array includes more than six subarrays.

In some embodiments, the plurality of photovoltaic modules 10 is installed directly to the roof deck 90. In some embodiments, each of the plurality of photovoltaic modules 10 is installed on the roof deck 90 by a plurality of fasteners. In some embodiments, the plurality of fasteners are installed through the headlap portion. In some embodiments, the plurality of fasteners includes a plurality of nails. In some embodiments, the plurality of fasteners includes a plurality of screws. In some embodiments, the plurality of fasteners includes a plurality of rivets. In some embodiments, the plurality of fasteners includes a plurality of staples.

In some embodiments, each of the plurality of photovoltaic modules 10 is installed on the roof deck 90 by an adhesive. In some embodiments, the adhesive is adhered

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directly to the roof deck 90. In some embodiments, the adhesive is adhered to an underlayment. In some embodiments, the underlayment is adhered directly to the roof deck 90. In some embodiments, the adhesive is located on a rear surface of the photovoltaic module 10. In some embodiments, the adhesive includes at least one adhesive strip. In some embodiments, the adhesive includes a plurality of adhesive strips. In some embodiments, the plurality of adhesive strips is arranged intermittently. In some embodiments, the adhesive is located proximate to one edge of the photovoltaic module 10. In some embodiments, the adhesive is a peel and stick film sheet. In some embodiments, the peel and stick film sheet includes at least one sheet of film removably attached to the rear surface. In some embodiments, the peel and stick film sheet is composed of Ever-Guard Freedom HW peel and stick membrane manufactured by GAF. In some embodiments, the adhesive includes polyvinyl butyrate, acrylic, silicone, or polycarbonate. In some embodiments, the adhesive includes pressure sensitive adhesives.

In some embodiments, one or more of the photovoltaic modules 10 is a photovoltaic module shown and described in either or both of U.S. application Ser. No. 17/831,307, filed Jun. 2, 2022, titled “Roofing Module System,” and published under U.S. Patent Application Publication No. 2022-0393637 on Dec. 8, 2022; and/or U.S. application Ser. No. 18/169,718, filed Feb. 15, 2023, titled “Roofing Module System,” and published under U.S. Patent Application Publication No. 2023-0203815 on Jun. 29, 2023, the disclosures of each of which are incorporated by reference herein in their entirety. In some embodiments, the photovoltaic module includes a structure, composition, components, and/or function similar to those of one or more embodiments of the photovoltaic modules disclosed in PCT International Patent Publication No. WO 2022/051593, Application No. PCT/US2021/049017, published Mar. 10, 2022, entitled “Building Integrated Photovoltaic System,” owned by GAF Energy LLC, and U.S. Pat. No. 11,251,744 to Bunea et al., issued Feb. 15, 2022, entitled “Photovoltaic Shingles and Methods of Installing Same,” the contents of each of which are incorporated by reference herein in their entirety.

In some embodiments, the system includes a plurality of roofing shingles. In some embodiments, the photovoltaic modules 10 have an appearance that aesthetically match the appearance of the plurality of roofing shingles. In some embodiments, the roofing shingle is a cuttable roofing module. In some embodiments, the roofing shingle is a nailable roofing module. In some embodiments, the roofing shingle is a cuttable roofing module shown and described in U.S. application Ser. No. 17/831,307, filed Jun. 2, 2022, titled “Roofing Module System,” and published under U.S. Patent Application Publication No. 2022-0393637 on Dec. 8, 2022; and/or U.S. application Ser. No. 18/169,718, filed Feb. 15, 2023, titled “Roofing Module System,” and published under U.S. Patent Application Publication No. 2023-0203815 on Jun. 29, 2023, the disclosures of each of which are incorporated by reference herein in their entirety.

What is claimed is:

1. A system, comprising:

a plurality of photovoltaic modules installed on a roof deck,

wherein each of the photovoltaic modules includes at least one solar cell,

a first encapsulant encapsulating the at least one solar cell,

wherein the first encapsulant includes a first surface and a second surface opposite the first surface,

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a frontsheet juxtaposed with the first surface of the first encapsulant,
 wherein the frontsheet includes a glass layer, and
 a backsheet juxtaposed with the second surface of the first encapsulant,
 wherein the backsheet includes
 a first layer,
 wherein the first layer includes a first edge, a second edge opposite the first edge, a third edge extending from the first edge to the second edge, and a fourth edge opposite the third edge and extending from the first edge to the second edge,
 wherein the first layer includes a first length extending from the first edge to the second edge, and
 a second layer adjacent and below the first layer,
 wherein the second layer includes a first edge, a second edge opposite the first edge of the second layer, a third edge extending from the first edge of the second layer to the second edge of the second layer, and a fourth edge opposite the third edge of the second layer and extending from the first edge of the second layer to the second edge of the second layer,
 wherein the second layer includes a second length extending from the first edge of the second layer to the second edge of the second layer, and wherein the first length is greater than the second length,
 wherein the second layer is the lowermost layer of the photovoltaic module,
 wherein the first edge of the first layer is offset and extends outwardly from the first edge of the second layer,
 wherein the second edge of the first layer is offset and extends outwardly from the second edge of the second layer,
 wherein the third edge of the first layer is offset from the third edge of the second layer, and
 wherein the fourth edge of the first layer is offset from the fourth edge of the second layer, and
 an adhesive layer between the first layer and the second layer,
 wherein the adhesive layer includes a first edge and a second edge opposite the first edge of the adhesive layer,
 wherein the adhesive layer includes a third length extending from the first edge of the adhesive layer to the second edge of the adhesive layer, and
 wherein the third length is equal to the second length.

2. The system of claim 1, wherein the first layer includes a first width extending from the third edge of the first layer to the fourth edge of the first layer, wherein the second layer includes a second width extending from the third edge of the second layer to the fourth edge of the second layer, and wherein the first width is greater than the second width.

3. The system of claim 2, wherein the second layer is composed of thermoplastic polyolefin (TPO).

4. The system of claim 2, wherein the first layer is composed of a polymer.

5. The system of claim 4, wherein the first layer is composed of thermoplastic polyolefin (TPO).

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6. The system of claim 2, wherein the glass layer includes:
 a first edge,
 a second edge opposite the first edge of the glass layer,
 a third edge extending from the first edge of the glass layer to the second edge of the glass layer, and
 a fourth edge opposite the third edge of the glass layer and extending from the first edge of the glass layer to the second edge of the glass layer,
 wherein the first edge of the glass layer is offset from the first edge of the second layer,
 wherein the second edge of the glass layer is offset from the second edge of the second layer,
 wherein the third edge of the glass layer is offset from the third edge of the second layer, and
 wherein the fourth edge of the glass layer is offset from the fourth edge of the second layer.

7. The system of claim 1, wherein the first edge of the first layer is offset from the first edge of the second layer by a first distance, wherein the second edge of the first layer is offset from the second edge of the second layer by a second distance, wherein the third edge of the first layer is offset from the third edge of the second layer by a third distance, and wherein the fourth edge of the first layer is offset from the fourth edge of the second layer by a fourth distance.

8. The system of claim 7, wherein the first distance, the second distance, the third distance and the fourth distance are equal to one another.

9. The system of claim 7, wherein the first distance is equal to the second distance, and wherein the third distance is equal to the fourth distance.

10. The system of claim 1, wherein the frontsheet includes a polymer layer and a second encapsulant, and wherein the polymer layer is attached to the glass layer by the second encapsulant.

11. A photovoltaic module, comprising:
 at least one solar cell,
 a first encapsulant encapsulating the at least one solar cell,
 wherein the first encapsulant includes a first surface and a second surface opposite the first surface,
 a frontsheet juxtaposed with the first surface of the first encapsulant,
 wherein the frontsheet includes a glass layer, and
 a backsheet juxtaposed with the second surface of the first encapsulant,
 wherein the backsheet includes
 a first layer,
 wherein the first layer includes a first edge, a second edge opposite the first edge, a third edge extending from the first edge to the second edge, and a fourth edge opposite the third edge and extending from the first edge to the second edge,
 wherein the first layer includes a first length extending from the first edge to the second edge, and a second layer attached to and below the first layer,
 wherein the second layer includes a first edge, a second edge opposite the first edge of the second layer, a third edge extending from the first edge of the second layer to the second edge of the second layer, and a fourth edge opposite the third edge of the second layer and extending from the first edge of the second layer to the second edge of the second layer,
 wherein the second layer includes a second length extending from the first edge of the second layer

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to the second edge of the second layer, and
 wherein the first length is greater than the
 second length,
 wherein the second layer is the lowermost layer of
 the photovoltaic module, 5
 wherein the first edge of the first layer is offset and
 extends outwardly from the first edge of the sec-
 ond layer,
 wherein the second edge of the first layer is offset
 and extends outwardly from the second edge of 10
 the second layer,
 wherein the third edge of the first layer is offset from
 the third edge of the second layer, and
 wherein the fourth edge of the first layer is offset
 from the fourth edge of the second layer,
 wherein the second layer forms a lower surface of 15
 the photovoltaic module, and
 wherein at least one of the first layer and the second
 layer is composed of thermoplastic polyolefin
 (TPO), and
 an adhesive layer between the first layer and the 20
 second layer,
 wherein the adhesive layer includes a first edge
 and a second edge opposite the first edge of the
 adhesive layer,
 wherein the adhesive layer includes a third length 25
 extending from the first edge of the adhesive
 layer to the second edge of the adhesive layer,
 and
 wherein the third length is equal to the second
 length. 30

12. A system, comprising:
 a plurality of photovoltaic modules installed on a roof
 deck,
 wherein each of the photovoltaic modules includes 35
 at least one solar cell,
 a first encapsulant encapsulating the at least one solar
 cell,
 wherein the first encapsulant includes a first surface
 and a second surface opposite the first surface,
 a frontsheet juxtaposed with the first surface of the first 40
 encapsulant, and
 a backsheet juxtaposed with the second surface of the first
 encapsulant,
 wherein the backsheet includes
 a first layer, 45
 wherein the first layer includes a first edge and a
 second edge opposite the first edge,
 wherein the first layer includes a first length
 extending from the first edge to the second
 edge, and a second layer adjacent and below the 50
 first layer,

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wherein the second layer includes a first edge and
 a second edge opposite the first edge of the
 second layer,
 wherein the second layer includes a second length
 extending from the first edge of the second layer
 to the second edge of the second layer, and
 wherein the first length is greater than the
 second length,
 wherein the second layer is the lowermost layer of
 the photovoltaic module,
 wherein the first edge of the first layer is offset and
 extends outwardly from the first edge of the
 second layer, and
 wherein the second edge of the first layer is offset
 and extends outwardly from the second edge of
 the second layer, and
 an adhesive layer between the first layer and the
 second layer,
 wherein the adhesive layer includes a first edge and
 a second edge opposite the first edge of the
 adhesive layer,
 wherein the adhesive layer includes a third length
 extending from the first edge of the adhesive layer
 to the second edge of the adhesive layer, and
 wherein the third length is equal to the second length.

13. The system of claim **12**, wherein the second layer is
 composed of thermoplastic polyolefin (TPO).
14. The system of claim **12**, wherein the first layer is
 composed of a polymer.
15. The system of claim **14**, wherein the first layer is
 composed of thermoplastic polyolefin (TPO).
16. The system of claim **12**, wherein the frontsheet
 includes a glass layer.
17. The system of claim **16**, wherein the frontsheet
 includes a polymer layer and a second encapsulant, and
 wherein the polymer layer is attached to the glass layer by
 the second encapsulant.
18. The system of claim **16**, wherein the glass layer
 includes a first edge, and a second edge opposite the first
 edge of the glass layer, wherein the first edge of the glass
 layer is offset from the first edge of the second layer, and
 wherein the second edge of the glass layer is offset from the
 second edge of the second layer.
19. The system of claim **18**, wherein the glass layer
 includes a third edge and a fourth edge opposite the third
 edge of the glass layer, wherein the third edge of the glass
 layer is offset from the third edge of the second layer, and
 wherein the fourth edge of the glass layer is offset from the
 fourth edge of the second layer.

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